

Raising Funds on Performance: Are Private Equity Returns Too Good to Be True?

Niklas Hüther*
Indiana University

March 19, 2018

Abstract

Recent studies on agency problems in private equity have fueled the suspicion that general partners (GPs) strategically manipulate performance estimates around fundraising times. While these studies rely on aggregated portfolio data, this paper offers the first empirical analysis of “window dressing” in private equity based on quarterly reported deal-level performance. Using information on buyout funds, I do not find any red flags of inflated performance on the deal-level. Observed interim peaks in valuation on the fund level result from timing of fundraising decisions to true estimates of high current asset values, while GPs cannot systematically repeat good outcomes.

*I am very grateful for valuable advice from David Robinson, Per Strömberg, Alon Brav, Lukas Schmid, Michael Ewens, Ludovic Phalippou, Merih Sevilir, Matthias Kahl, Adriano Rampini, Roberto Steri and Jonathan Zandberg. I also thank an anonymous investor and seminar participants at Duke University, Indiana University, Erasmus University Rotterdam, University of Nebraska—Lincoln, Southern Methodist University, as well as seminar participants at the 2016 European Finance Association Meetings, the 2016 PERC conference, and the 2017 Conference on Entrepreneurial Financial Management at ESMT Berlin. Address: Kelley School of Business, Indiana University, 1275 E 10th St, Bloomington, IN 47405. Phone: +1 919-808-6829; Email: nhuether@indiana.edu.

1 Introduction

Investments in private equity are typically structured as ten year limited partnerships in which fund managers act as general partners (GPs) and investors act as limited partners (LPs). Since the fund life is broken down into an investment and a liquidation period, GPs can only make new investments after the investment period has expired by raising a new fund. At that time, existing investments are not necessarily liquidated, so that current fund returns rely heavily on subjective performance estimates of their investments. This fact, stemming from a market setting of information asymmetries, has led many investors, industry observers, and academics to speculate that private equity firms distort their performance measurement around fundraising events. While this has been widely alleged, data limitations have made it difficult to draw sharp conclusions about this concern.

Various prior studies have focused on agency problems around fundraising, centering on the question of whether agents inflate portfolio values prior to fundraising. The collection of evidence indicates that performance estimates are inflated particularly for investments in low reputation funds (e.g., Barber and Yasuda 2017, Chakraborty and Ewens 2017 and Brown et al. 2017).¹ Results by Barber and Yasuda (2017) and Brown et al. (2017) suggest that less reputable funds face lower costs of manipulation and are therefore more likely to strategically inflate valuation estimates compared to high reputation funds. This result is in line with Chakraborty and Ewens (2017) who find increases in valuation before fundraising among low reputation VC firms, as well as an increase in write-offs after fundraising. High net asset values (NAVs) during fundraising for both low and high reputation funds are reported by Jenkinson et al. (2013).

¹Investing in private equity requires investing in low reputation funds, at least initially, to gain a “seat at the table” once their GPs become more reputable (see e.g., Lerner, Schoar and Wongsunwai 2007).

The common finding of these studies is that fundraising for a new fund coincides with times of high current interim valuations, especially when costs of manipulation appear low. This finding is open to two different interpretations. While GPs may advertise strong current fund performance by manipulating true estimates of current asset values as suggested by prior work, alternatively, GPs may time fundraising around true estimates of high current NAVs.

Motivated by prior research, I argue that the data underlying these studies are aggregated too coarsely to attribute performance peaks to inflation of underlying asset values. To overcome data limitations of prior work, I have built a novel database of quarterly deal valuations in U.S. buyout funds that allows me to address the hypotheses above. To the best of my knowledge, this is the first study that utilizes individual portfolio company data of buyout funds to investigate the hypothesis of window dressing due to agency frictions around fundraising.²

For a better understanding of why it is important to have individual portfolio company data, consider a fund with heterogeneous investment times. A peak of net asset values on the fund level, for instance, could be explained by inflated interim valuations of individual portfolio holdings. Alternatively, if deals made shortly before fundraising perform poorly so that their valuations after fundraising fall below their initial valuations at cost, these deals could reduce portfolio valuations after fundraising. If, in addition deals made long before fundraising perform well, so that their initial valuations at cost increase around fundraising, the result would be a peak in aggregated values on the fund level just before a new fund is raised. Typically, investments are valued at cost when they enter the books (see, e.g., Barber and Yasuda 2017 and Chakraborty and Ewens 2017).³

²Typically, buyout funds provide more scope for manipulation as compared to venture capital funds, since valuations are not tied to a per share price that a VC investor is willing to pay in a financing round.

³I find that both well-performing deals as well as write-offs are, on average, valued at cost in the first two quarters after the initial investment. See also Figure 2.

One set of my results speaks directly to these two possible explanations: that is, whether high interim valuations on the fund level can be explained by inflated interim deal valuations, or whether they are a result of deal composition. Indeed, the return of a deal between fundraising and exit is on average not significantly lower in funds that lack reputation and may anticipate problems in fundraising. I also find no signs of systematically higher interim deal valuations for other measures of anticipated problems in fundraising, such as prior unsuccessful deals. Thus, this evidence is inconsistent with GPs strategically manipulating NAV estimates, but leaves room for a deal composition explanation and timing of fundraising to true high values of current portfolio holdings. Consistent with the deal composition and timing explanation, I find strong evidence that low reputation funds that try to raise a new fund have, on average, more successful investments well ahead of fundraising compared to deals made shortly before fundraising. Realized (or last observed) value multiples (VMs) of investments made long before the fundraising event exceed multiples of investments made shortly before the event, on average, by 130 percentage points. The difference stems from VMs of investments by low reputation funds that were unrealized at the time of raising a new fund. This indicates that peaks in performance shortly before fundraising are a result of deal composition instead of inflated NAVs.

Despite finding no evidence of performance peaks on the deal level, less reputable GPs that try to fundraise could still hasten or delay mark ups of deal values strategically around fundraising. To test for this possibility, I compare the valuation of the same deal across two GPs that hold this deal in their portfolio, where one is fundraising and the other one is not. Mirroring previous results, I find no systematic differences in valuations of the same deal across different GPs.

Thus, the conjecture about manipulation is inconsistent with the deal-level evidence on fundraising. The deal composition result of strong early deals increasing in value well ahead of fundraising and weak later investments declining in value primarily after

fundraising is, instead, consistent with a timing of fundraising story. This finding, however, prompts two additional questions: (a) Do these strong early investments by low reputation funds reflect skill or luck? and (b) Are poor investments made shortly before fundraising a result of opportunistic investment behavior rather than fund timing? From an agency perspective these questions are important, since they leave room for agency frictions even if GPs do not strategically inflate NAVs.

To address the first question, I document and exploit the fact that, on average, 10% of a private equity fund's portfolio is not private equity but rather represents investments in publicly traded stocks that are not intended to be taken private. I observe Schedule 13D filings accompanied with these investments, regardless of whether they are PIPEs or investments on the public stock exchange. My strategy of identifying investor skill is based on the market's reaction to the announcement of the investor's identity. I find large positive average abnormal returns on the filing day, of 2% (4%) for low (high) reputation funds. I also find that the positive returns at announcement are not reversed over time. If low reputation funds were purely lucky in "putting some successful early points on the board", the market should not react to their investments—or at least those abnormal returns should be later reversed. Under the assumption that private and public investments are expected to generate similar returns to justify both types of investments to LPs, this finding indicates that there is some level of skill involved when choosing investments.

Both timing of fundraising decisions and opportunistic investment behavior can explain why good investments are found well ahead of fundraising and poor investments occur shortly before fundraising. The idea behind opportunistic investment behavior is that low reputation funds may lack the network among private equity firms or the deal flow and thus need more time in finding good investment opportunities compared to high reputation funds. If GPs need more time finding good investments early on, they may face greater pressure to deploy unused capital towards the end of the contractual

investment period.⁴ According to Arcot et al. (2015), private equity funds that are under pressure pay a premium for their investments, which would explain the underperformance of investments made shortly before fundraising. To test for this possibility, I compare dry powder across funds shortly before fundraising and find no statistical and economic difference between low and high reputation funds.

From another perspective, if low reputation funds behaved opportunistically by making worse investment choices later in the investment period, as mentioned above, they should do so unconditionally, whether they ex post try to raise a new fund or not. My data allows me to differentiate between attempted and unattempted fundraisers based on whether due diligence material was sent out to the investor during the current fund life. Inconsistent with the presumption that low reputation funds face higher pressure to deploy dry powder at the end of the investment period, I find no evidence that early deals perform better than later deals among the subsample of unattempted fundraisers. That said, my results show that VMs of investments made in the first two years of the life of attempted fundraisers are, on average, significantly above VMs of deals thereafter. This result is consistent with general partners raising a new fund after experiencing good outcomes which low reputation funds cannot systematically repeat.

To actually test whether GPs are more likely to attempt to raise a new fund following true good performance estimates I extend the result by Barber and Yasuda (2017). Their result suggests that NAV window dressing predicts fundraising, and this effect is concentrated among those who most need to rely on interim NAVs as stand-ins for reputation. By using hazard rate models, I find that fundraising is endogenously timed to true estimates of high current valuations, while their performance, especially in low reputation funds, cannot be sustained in later deals. Consistent with this evidence, GPs appear to verify strong private performance with investments in publicly traded stocks (PIPEs or investments on the public stock exchange that are not taken private).

⁴Typically a private equity fund can only make new investments within the first five years of the fund's life.

High positive returns in these public deals have a significant impact on the probability of attempting to raise a follow-on fund. In contrast to endogenous exits, which are found to make fundraising more likely (Barber and Yasuda 2017), returns of these stock investments are publicly observable throughout their holding periods. This finding is consistent with Chakraborty and Ewens (2017) who claim that the quickening of exits by GPs can lead LPs to pass on investing in the next fund and should rarely be used by PE firms. Overall, my findings are consistent with the model proposed by Shin (2003), showing that it is optimal for a manager to disclose the best possible outcome to interested parties by verifiable disclosures, since an effective penalty would always annihilate reporting false evidence.

The rest of the paper proceeds as follows. Section 2 discusses the sample. Section 3 confirms existing evidence of performance peaks on the fund level around fundraising times. Section 4 explores whether net asset values of underlying investments are strategically manipulated and considers what explains performance peaks on the fund level. Section 5 investigates whether GPs time fundraising to true estimates of high current asset values, and explores how public deal performance impacts the probability of raising a follow-on fund. Section 6 concludes.

2 Data

The data in my study were provided to me by one of the largest international LPs in the world on an anonymous and confidential basis. Although the data source is a large, global investor who invests in various private equity asset classes, I restrict my analysis to U.S. buyout funds to narrow the scope of investment focus with regard to the research question. The data set comprises 2,776 fund-investment pairs of 136 funds with vintage years between 1996 and 2010. About 40% of exited investments are realized below cost

and 17% are completely written off.⁵ While 19.3% of the sample funds' investments were publicly traded at some point during their holding period, 7.7% are fund investments in publicly traded stocks that remained publicly traded at least until their exit. These fund-investment pairs correspond to 2,400 unique sample portfolio companies, of which 16.1% are (at some point) publicly held in the fund's portfolio, and 8.4% are investments in publicly traded stocks that are not taken private. Being able to differentiate between investments held by several funds, I can identify 239 (127) privately held companies in the portfolios of at least two different funds (at least two different GPs).⁶

Of the 136 funds, 121 engaged in marketing to raise a new fund, whereas 15 did not try to advertise a follow-on fund—by sending out due diligence information to the institutional investor—before December 31, 2016. Since funds with vintage years beyond 2010 are not included in the sample, these 15 funds are considered unattempted fundraisers.⁷

I also do not include funds in my sample that raised a new fund after December 31, 2013, the last date for which I observe NAVs and cash-flows. For the remaining 121 funds, I have access to the date of the reported performance in the due diligence package that was sent out to advertise a new fund. Considering that existing fund investors still observe performance after the report date of the current fund's performance in the due diligence package, I define the fundraising event for these 121 sample funds as the first quarter after closing (expected closing) of the next fund.⁸

⁵This is consistent with Lopez-de-Silanes et al. (2015) who report 15% of complete write-offs in their sample.

⁶Privately held means deals that were privately held at some point in time in the fund's portfolio.

⁷According to Barber and Yasuda (2017), PE firms typically manage to raise a new fund by the sixth year of the current fund's life.

⁸For funds that advertised a new fund but ultimately did not fundraise, I use the expected first closing from the PPMs/due diligence documents.

2.1 Sample Representativeness and Basic Summary Statistics

I assess the representativeness of my sample by comparing it with publicly available data collected by Thomson One and the other related studies.

—Please see Table 1—

While Table 1 shows that my sample is smaller compared to the number of funds and portfolio companies in Thomson One, my data allow me to analyze reported NAVs and cash flows on the individual portfolio company level. This is a big advantage, as normally GPs transmit only quarterly net-of-fees fund level performance (see, e.g., Metrick and Yasuda 2010, Robinson and Sensoy 2013). In addition, I observe the GICS industry group for each portfolio company.

To identify funds with a potential strong incentive to strategically manipulate valuations, I split the sample into high reputation and low reputation funds. Low reputation funds must place more emphasis on convincing prospective LPs that they are skilled. These funds lack a strong past track record and need more time to reach a first close compared to vintage year cohorts. Thus, I define low reputation funds as having no prior top quartile performing funds that are more than five years old from the inception of the sample fund (low performance) and where the time to fundraise from sending out due diligence material to first closing of the sample funds is above the sample median.⁹

Descriptive statistics of the funds and their GPs are reported at the bottom of Table 1. The average (median) vintage year in my sample is 2004 (2005), which is comparable to Thomson One. However, I find that my sample consists of statistically and economically larger funds that are raised by more established GPs than the average and median counterparts from Thompson One. This is partly attributable to the fact that

⁹As a result of multiple conversations with different GPs and LPs, the time to fundraise in comparison to vintage year cohorts is claimed to be a strong indicator of GPs' reputation. (This is also in line with reports by Preqin.) The low performance definition is in line with Barber and Yasuda (2017).

the large size of the Investor in question precluded them from investing in small funds. However, if compared to other recent studies with fund level NAV data, for example, Barber and Yasuda (2017) and Brown et al. (2017), I find a smaller gap in the size of my analyzed funds (3,107 million USD vs. 1,532 million USD, as in Barber and Yasuda (2017), or 1,324 million USD as in Brown et al. (2017) vs. 579 million USD, as in Thomson One). Recent studies such as Metrick and Yasuda (2010), also using data from a large independent LP, report similar average fund sizes (1,238 million USD) to those of Barber and Yasuda (2017) and Brown et al. (2017). Considering that Metrick and Yasuda (2010) have data until 2006, and thus leave out the heydays of fundraising in the U.S. buyout market between 2006 and 2008, my average fund size seems to be comparable.

In relation to Thomson One's data universe, my GPs are, on average, older and larger in terms of previous investment activity. To capture this, I compute the size of previous funds operated by the same GP as a fraction of the total investment activity in the sector over the previous ten years. My sample GPs appear more experienced as active investors, rather than displaying "zombie" firms with an average of 5 funds for a life time of 15 years versus an average of 2 funds in 11 years, as in Thomson One.

A data set of generally more experienced GPs might raise concerns that these GPs should have a low incentive to strategically manipulate NAVs. However, a fraction of 26% of these funds did not raise a follow-on fund.¹⁰ This fraction is similar to related studies (31% as in Chakraborty and Evens 2017, 30% as in Barber and Yasuda 2017, and 14% as in Brown et al. 2017). Brown et al. (2017) find that more experienced GPs are more likely to have a follow-on fund, although the relationship is not monotonic. A typical example, discussed in the media, is the failed fundraising attempt of the fourth U.S. buyout fund by J.W. Childs Associates in 2007. The private equity firm was in

¹⁰These are funds that have not been able to raise a follow-on fund for at least 6 years since initiation (as of 12/31/2016 with 2010 as the last vintage year in my sample). For the subsample of funds that advertised a new fund 17% failed in their attempt to fundraise (funds without a first closing *at least* 3 years after due diligence information was sent out).

business for 12 years at that time and had around 3 billion USD under management.

2.2 Types of Investments and Detailed Summary Statistics

Turning to portfolio investments with publicly observable valuations, I distinguish between different types of transactions, that is, IPO exits, public-to-private / public-to-private-to-public deals, and investments in publicly traded stocks (not taken private). While the latter has received little attention in the prior literature so far, Table 2 shows that investments in publicly traded stocks that are not taken private before exit account, on average, for 8% (11% in terms of invested capital) of a private equity portfolio, equaling the portion of IPO exits. As mentioned in the introduction, in contrast to endogenous IPO exits, returns of these stock investments are publicly observable throughout their holding period.

—Please see Table 2—

Table 2 reveals that about 60% of the 136 U.S. buyout funds hold such public stock investments, and it breaks the funds' fractions down into different types of publicly held deals (as defined above) by number and dollar amount (in brackets). Next to that, we can see the average number of portfolio companies. Numbers are stated for the entire sample, as well as subsamples of attempted and unattempted fundraisers.

On average, funds hold 20 companies in their portfolio during the sample fund life (SFL), except for low reputation funds with slightly fewer investments, and around 15 at a fundraising event (FRE). Compared to publicly held investments and IPO exits which comprise the largest portion of public deals in the sample funds, a lower portfolio fraction constitutes of going-private transactions (on average around 3% or 5% of invested capital). Deals that were taken private and afterwards taken public again represent the smallest fraction of a private equity fund's portfolio (on average below 1%). Averages

remain similar across the subsamples of fundraisers, and we can see that the majority of investments are made into private companies that are not taken public (on average around 80% of the portfolio companies or 70% of the fund's invested capital). That being said, Table 2 points out the economic relevance of funds' investments with publicly observable valuations—also around a fundraising event— and specifically portfolio allocations to publicly traded companies with verifiable NAVs during their entire holding time.

—Please see Table 3—

Statistics in Table 3 show the distribution of funds of the 121 attempted fundraisers in my sample—which are the focus of my multivariate analysis—and their publicly held investments over time and across industries. Observations are on the fund-deal pair level. Panel A shows that apart from going-private followed by going-public transactions, publicly held investments generally span the entire sample period, with peaks in numbers around the late 1990s / early 2000s and from 2005-2008. Regarding distributions across industries based on the GICS industry classifications that are displayed in Panel B, I find the majority of companies within the consumer discretionary, industrial, and information technology sector (in line with Braun et al. 2017). Distributions are similar across different types of deals.

2.3 Average Deal Performance by Investment Year

In Table 4, I provide descriptive statistics on VMs of fund-deal pairs by year of initial investment for the 121 attempted fundraisers.¹¹ I report VMs based on the full sample and differentiate between realized and unrealized investments. Unrealized investments

¹¹Results based on IRRs are qualitatively the same and can be found in the appendix. According to GPs and LPs, the industry typically focuses on the VM rather than the IRR, because investors tend to be more concerned about obtaining higher total equity returns than about receiving the investment back in a shorter time period (that is “juicing” the IRR of the investment).

are investments that are still in the fund's portfolio at the end of the sample period as of 12/31/2013, with valuations based on reported NAVs.

—Please see Table 4—

For the 2,451 fund-deal pairs, the mean (median) VM is 1.83 (1.4). For the subsample of realized deals, the mean (median) is 2.09 (1.61), while for unrealized deals it is 1.53 (1.11). These numbers parallel those from a sample by Braun et al. (2017), finding a median VM of 1.5 for their full sample, 1.9 for realized deals, and 1.0 for unrealized deals. Meanwhile, Lopez-de-Silanes et al. (2015) document a median of 1.7 for a slightly older sample period.¹²

Turning to the distribution of deal performance over initial investment years, Table 4 shows the procyclicality of private equity investments—those that are made in up-markets—and the countercyclicality of private equity returns. While investments made between 1996 and 1998 did quite well, those initiated between 1999 and 2001 exhibited low VMs. This trend is reversed afterwards. In particular, investments that were initiated between 2002 and 2005 experienced high realized VMs given their exits during the heydays of buyout funds and the valuation effect of money chasing deals (Gompers and Lerner 2000). Investments made during the financial crisis exhibited high returns as well. The finding of procyclical investment patterns (column 2) and divestment decisions is in line with Robinson and Sensoy (2016). Additionally, the realization status reveals that investments made in the mid-1990s still had some unrealized deals in 2013, while returns from recent investments have not been harvested two years later, at the end of the sample period (in line with Braun et al. 2017). The consistency with descriptive statistics of comparable samples from other studies is reassuring. In the remainder of this analysis, I will make use of my access to reported asset valuations, especially around fundraising, and focus on the question of performance manipulation.

¹²The unique feature of my dataset—relevant for my research question—is that I observe valuations of deals that are reported *quarterly during their holding period*.

3 Reported Fund Performance

In line with the perception that especially low reputation GPs have an incentive to display strong performance around a new fundraising event, Figure 1 shows that cumulative abnormal fund returns peak around the time a new fund is raised for the subsample of low reputation funds. This result has also been reported in previous work (Barber and Yasuda 2017; Brown et al. 2017; Chakraborty and Evens 2017; Jenkinson et al. 2013). Both graphs in Figure 1 display average cumulative excess returns over the S&P500,

$$CAR_t^{NAV} = \prod_{s=1}^t (1 + r_s^{NAV}) - \prod_{s=1}^t (1 + r_s^m), \quad (1)$$

where t and s index the period in event time, r_s^{NAV} is the fund's return in event quarter s , i.e., over the time interval from $s - 1$ to s , and r_s^m is the market benchmark return in event quarter s . The fund's quarterly reported return is defined as

$$r_s^{NAV} = \frac{NAV_s - (NAV_{s-1} + C_s - D_s)}{NAV_{s-1}}. \quad (2)$$

While the graph on the left sorts fund quarters chronologically, the one on the right sorts quarters in event time (by the time to and from fundraising).

Besides the visual presentation of fund performance peaks over time, Table 5 looks at the cross-sectional post-fundraising performance of attempted fundraisers. Since reputation has been shown to mitigate agency issues related to fundraising (Barber and Yasuda 2017 and Brown et al. 2017), Table 5 presents performance differences between low and high reputation funds. To do so, I isolate the performance after fundraising by calculating fundraising value multiples (FVMs) that have also been used by Barber and Yasuda (2017). The FVM is gross-of-fees and assumes an investor invests in an existing fund at the first quarter after closing (expected closing), that is, at the fundraising event (FR), and holds the fund until liquidation or until the end of the sample period

(31/12/2013), whichever occurs earlier:

$$FVM_{iFR} = \frac{\sum_{\tau=FR+1}^T D_{i\tau} + NAV_{iT}}{NAV_{iFR} + \sum_{\tau=FR+1}^T C_{i\tau}}, \quad (3)$$

where $D_{i\tau}$ and $C_{i\tau}$ are the cumulative distributions from investments and invested capital, respectively, for fund i in quarter τ and T is the fund's liquidation quarter / last quarter in the sample period.

—Please see Table 5—

Column (1) of Table 5 presents the baseline regression estimates of regressing FVMs on a low reputation dummy variable that takes the value of one in the case of a low reputation fund, and zero otherwise. To control for heterogeneity in market conditions and fund life, I include calendar year and fund quarter fixed effects in all models. The average post-fundraising performance for low reputation funds is significantly lower by on average 31 percentage points. This result suggests that less reputable GPs systematically inflate their NAVs and defer their write-offs until after the fundraising event, which is in line with prior work.

Column (2) introduces a dummy variable for failed public deals, pre-fundraising. This variable is negatively correlated with post-fundraising performance, which again is consistent with the speculation that GPs inflate their NAVs if they anticipate problems in fundraising. That said, interacting low reputation funds and prior unsuccessful public deals in column (3) does not show that low reputation funds holding public deals with large negative returns (e.g., Washington Mutual), as a measure of anticipated problems in fundraising, experience a stronger decline in post-fundraising performance as compared to funds without a large negative return in public deals. By looking at unsuccessful prior exits (columns 4), we can see that the sign of the coefficient is also negative, while there seems to be a loss of statistical power (see also column 5). This could be because funds tend to delay write-offs until fundraising is completed (Chakraborty and

Ewens 2017). The mechanism that is at play for the observed performance peaks in the subsample of low reputation funds remains unclear and gives reason to speculate—as in previous literature—that NAVs are strategically inflated.

4 Understanding Reported Fundraising Performance

4.1 Interim Deal Valuations

To get a better traction on this speculation, I apply multiple analyses by looking into the reported fund level NAVs and using individual NAVs of their holdings. I begin with Table 6 by exploring FVMs, introduced in Section 3, on the deal level. Overall, the results indicate that low reputation funds do not statistically differ in their FVMs compared to high reputation funds. That is, I find no evidence that these funds inflate interim NAVs of their portfolio holdings in contrast to results of aggregated fund returns.

—Please see Table 6—

Column (1) only includes the low reputation fund dummy variable that shows a negative loading but no statistical significance. There is no significant relation between additional measures for anticipated problems in fundraising—that is, unsuccessful public deals or unsuccessful exits prior to fundraising. Larger deals show no statistically significant difference in investment performance after fundraising. All models include calendar year, deal quarter, and industry fixed effects. As shown by this analysis, I find no evidence that performance peaks around fundraising stem from inflated interim investment NAVs, which to the best of my knowledge, could not be tested due to the lack individual portfolio company data.

4.2 Deal Composition

If performance does not appear to be inflated on the investment level, the reasoning for the seemingly inflated NAVs on the fund level still remains unclear. In that regard, Figure 2 compares cumulative returns of investments made long and shortly before fundraising and sheds more light on an alternative explanation for the signs of window dressing on the aggregated portfolio level. The left hand side of Panel A displays cumulative abnormal returns of investments made two and a half years before the fundraising event. After about the first two quarters, where GPs typically book their investments at cost, returns are increasing and are generally valued below their final exit value indicated by jumps in returns after about three years of holding. Conservatively reported NAVs before exit have also been documented by Barber and Yasuda (2017), Brown et al. (2017) and Jenkinson et al. (2013). Most importantly, average returns are nearly monotonically increasing until exit. Unlike with returns of investments made long before fundraising, which differ in a few strong exit values between low reputation and high reputation funds, there are substantial differences in returns of investments that occur shortly before fundraising, as shown on the right side of Panel A. While cumulative returns are increasing for high reputation funds after their initial valuations at cost, values of investments made shortly before fundraising fall below their initial valuation at cost for low reputation funds. Declining returns primarily post-fundraising in combination with strong deals well ahead of fundraising and increasing returns around fundraising appear to produce performance peaks on the aggregated fund level for low reputation funds.¹³

—Please see Table 7—

To dive deeper into this potential explanation based on deal composition, Table 7 looks at the difference in performance of deals made long and shortly before fundraising in

¹³Graphs are qualitatively similar for up to four years (up to one year) before fundraising.

a multivariate setting. Overall results for attempted fundraisers show that investments' VMs, on average, decrease by 0.47 for investments made shortly before a fundraising event compared to long before one, which is in line with the raw performance difference displayed in Figure 2. Only investments that remain unrealized for at least one year after the fundraising event are considered. The coefficient drops conditional on successful public deals, pre-fundraising (column 2). This result is at odds with the suspicion that GPs tend to engage in window dressing when anticipated problems in fundraising are more severe, while instead suggesting the importance of performance verifiability with respect to fundraising, an issue analyzed more in Section 5. With Figure 1 as a prelude, these negative loadings which are highly statistically significant stem from the subsample of low reputation funds (columns 5-6). All models include fixed effects for the deal life, the calendar year at fundraising and the industry.

4.3 Club Deals

While results indicate that performance peaks on the fund level are a result of deal composition rather than inflated interim deal valuations, they do not rule out the possibility that GPs book companies less conservatively around fundraising. In other words, GPs might quicken or delay mark ups of deal values strategically. To test for the possibility of strategic booking, I compare valuations of the same deal across at least two GPs (club deals), where one is fundraising and the other one is not. Ultimately, I compare valuations of 127 club deals in my sample. For these 127 club deals, I compare FVMs each time there is a fundraising event. Thus, my sample includes 443 observations where club deals are privately held and unrealized at a fundraising event.

To set the stage, Figure 3 shows the percentage of cases where the fundraising GPs have the lowest FMVs (left bar), the same FMVs as non-fundraising GPs (middle bar), and where non-fundraising GPs have the lowest post-fundraising quarter performance (right bar). In only one-third of all investments with the lowest post-fundraising perfor-

mance is the GP actually fundraising, while in nearly 60% of all cases the lowest FMVs are linked to non-fundraising GPs. The small percentage of investments with the same FMVs indicates that GPs do not seem to coordinate their valuations of the same deal.

To formally test whether fundraising GPs increase the valuation of their deals compared to other invested GPs, I regress FMVs in the same company on an event dummy that is equal to one if the investment is made by a fundraising GP in the event quarter, and zero otherwise.

—Please see Table 8—

Results presented in Table 8 show there is no statistically significant difference in post-fundraising performance between fundraising and non-fundraising GPs that are invested in the same deal, controlling for industry, calendar year, and deal quarter. Results remain qualitatively unchanged by considering subsamples of high and low reputation funds (columns 3-6). The combined interpretation of my portfolio company level results is that there is no evidence of strategically inflated valuations.

4.4 Analysis of Investment Skill and Opportunistic Behavior

As pointed out in Section 4.2, instead of inflated deal values low reputation funds appear to make bad deals shortly before fundraising that drop in valuations primarily after fundraising as well as good deals made well ahead of fundraising that increase in value around the event. This result calls into question whether agency conflicts are represented in these performance patterns. One concern is these successful early investments in low reputation funds display luck rather than skill. Making fund-commitment decisions based on returns of pure luck would put new investors at risk of chasing non-fundamental values.

Separating skill from luck is a difficult task. While a recent study by Korteweg and Sorensen (2017) uses a variance decomposition model to assess the skill of PE firms based on fund level performance, my approach is to identify investor skill by the market reaction to the announcement of the investor from a Schedule 13D filing. As mentioned in the introduction to this paper, about 10% of a private equity portfolio represents PIPES, or investments on the public stock exchange that are not taken private, and all such investments in my sample are accompanied by a Schedule 13D filing. In contrast to investments in publicly traded companies, private-to-public deals in my sample are accompanied by a Schedule 13G filing—indicating a passive investment. The announcement returns to a Schedule 13D filing / announcement of activism provides information about the market’s perception of investor skill.

Figure 4 plots the average buy-and-hold return, above the buy-and-hold return on the U.S. MSCI industry indices, in event time—20 days before and 20 days after the Schedule 13D filing day. Abnormal returns are plotted separately for low reputation funds (dashed line) and high reputation funds (solid line), with a run-up of about 4% and 2%, respectively, for high- and low-reputation funds at the filing day. This means a level of GP’s skill is reflected in the market reaction from taking a position in a publicly traded company. If GPs’ successful early investments were purely a reflection of luck, one would not expect to see an abnormal announcement return, or one would expect at least a reversal of such. Figure 4, however, does not show a reversal of abnormal returns 20 days after the announcement, and I also do not find that announcement returns are, on average, reversed until the investment’s exit date or until the end of the sample period, if they are still unrealized at that point.¹⁴ Under the assumption that private and public investments are expected to generate similar returns to justify both types of investments to LPs, this finding indicates that there is some level of skill involved when choosing investments.

¹⁴In unreported results, I calculated long-term buy-and-hold abnormal returns beginning 20 days after the Schedule 13D filing date and 20 days before this date. The average abnormal returns are greater than zero in both cases and statistically significant for the latter.

Another question is whether the timing of fundraising decisions or opportunistic investment behavior explains why good investments are found well ahead of fundraising and poor investments occur shortly before fundraising. The idea behind opportunistic investment behavior is that low reputation GPs may lack an established network and good deal flow so that they need more time to find promising investments early on compared to high reputation GPs, meaning they face stronger pressure to deploy unused capital at the end of the investment period. This hypothesis is motivated by Arcot et al. (2015), who find that funds buying firms under pressure tend to overpay.

Figure 5 reports the fraction of capital remaining in the current funds (dry powder) that attempted to raise a new fund. The average percentage of dry powder is displayed in event time and remains slightly higher for high reputation funds (solid red line) compared to low reputation funds (dashed blue line), while the percentages converge around fundraising. This univariate comparison is at odds with the presumption that low reputation funds hold a higher percentage of dry powder shortly before fundraising and thus feel strong pressure to deploy unused capital.

—Please see Table 9—

Table 9 examines this in greater detail with two-way and three-way interaction models. Models (1) to (4) show that, during the time shortly before a fundraising event, dry powder declines as expected. Model (2) introduces an interaction term of the shortly-before-fundraising indicator and the low reputation dummy that turns out to be statistically insignificant in line with Figure 5. To compare funds with a larger spread in anticipated problems in fundraising, I build three-way interaction models reported in columns 3 and 4. I basically interact the interaction term of model 2 with an indicator for unsuccessful public deals (Model 3) or unsuccessful exits (Model 4), pre-fundraising, while also including all two-way interaction terms. The three-way interactions in models (3) and (4) show that the two-way interaction remains without effect across differences in anticipated problems in fundraising.

From another perspective, if low reputation funds behaved opportunistically by making worse investment choices later in the investment period, they should do so unconditionally, whether they ex post try to raise a new fund or not. Access to due diligence information received by the investor allows me to identify 159 investments of low reputation sample funds that did not attempt to raise a follow-on fund. Equipped with this subsample, results of Table 10 show that only investments made in the first two fund life years by attempted fundraisers are significantly better performing compared to deals made later in the funds' investment period. VMs of these good early deals are on average 90 percentage points higher than VMs of investments made thereafter. This difference declines and is no longer statistically significant when combining the subsample of attempted and unattempted fundraisers.

—Please see Table 10—

Overall these results are consistent with the idea that investment success of low reputation funds is to some extent related to skill, and GPs raise a new fund after experiencing good outcomes which low reputation funds cannot systematically repeat. I find no evidence consistent with opportunistic investment behavior explaining these performance patterns.

5 Raising Funds on True Performance

To actually test whether GPs are more likely to attempt to raise a new fund following true good performance estimates I extend the result by Barber and Yasuda (2017). Their result suggests that NAV window dressing predicts fundraising, and this effect is concentrated among low reputation funds that lack good exits around fundraising and need to rely more on interim NAVs. Barber and Yasuda (2017) also find that good exits, representing verifiable transactions, tend to improve the probability of fundraising. This

finding is consistent with the grandstanding effect first documented by Gompers (1996).

There are two potential concerns with this interpretation. First, clustering exits early in the fund's life is an observable action that could lead to LPs passing on investing in the next fund (Chakraborty and Ewens 2017). Second, the finding that performance backed by strong realizations impacts fundraising decisions does not rule out reverse causality if realized returns are based on endogenous exits. This means the documented effect of performance on fundraising is ambiguous due to endogeneity concerns.

5.1 Returns of Stock Investments as Verifiable Performance

To address these concerns and the the finding that funds without good exits appear to lack verifiable returns and rely more on NAV window dressing, I make use of the fact that I observe publicly observable returns of stock investments in private equity funds by looking at deal level data (see also Section 2.2). In doing so, I provide evidence that GPs appear to use these stock investments as a way to back less verifiable valuations of unexited companies.

Figure 6 illustrates the timing of investments according to whether they are privately held or publicly traded. The left column of Figure 6 depicts the distribution of investment times for high reputation funds as a function of event time (by the time to and from fundraising), while the right column only includes low reputation funds. These graphs are generated by pooling all initial investments by fundraising quarter for private and public type deals and then plotting the distributions of investments. There is very little difference in the distribution of investment times between private and public deals of high reputation funds. The investment times are approximately normally distributed with the median around one year before the event, meaning that funds have made half of their investments a year before attempting to raise a new fund.

Unlike with deals of high reputation funds, there are substantial differences in the distribution of investment times between private and public deals of low reputation funds, as can be seen in the right column of Figure 6. For private deals, the mean/median investment occurs again around a year before the event, while the mean/median stock investment is made about two years before fundraising. These differences in means/medians are accompanied by substantial variations in distributions. It appears that low reputation funds try to increase their fundraising prospects by placing early investments in stocks to back up private performance with publicly observable returns. The finding is also consistent with the idea that early success is not due to later pressure to invest unused capital as discussed in Section 4.4. Stock investments are apparently not used as a tool to deploy dry powder quickly just prior to raising a new fund.

5.2 Fundraising Timed to Performance

To test whether GPs endogeneously time fundraising to true estimates of high performance that cannot be systematically repeated in later deals, especially for low reputation funds, I model the probability that a GP raises a new fund in a quarter as a function of time-varying performance. In addition, I test whether interim performance estimates have a greater impact on fundraising probability when they are backed by investments in publicly traded stocks. Along the lines of Barber and Yasuda (2017), an event is considered a failure in the hazard model setting if the GP raises a new fund during the current fund's life. The hazard rate specification of a fundraising event for fund i at fund quarter t is:

$$h(t|\mathbf{x}_{it}) = h_0(t)\exp(\mathbf{x}_{it}^T\boldsymbol{\beta}), \quad (4)$$

where $\mathbf{x}_{it}$ is a vector of fund characteristics (of which most are time dependent), $\boldsymbol{\beta}$ is a parameter vector, and $h_0(t)$ is the baseline hazard function.

Figure 7 displays the Kaplan/Meier survival function for the fundraising events of the 121 attempted fundraisers in my sample over fund life quarters 1-40. The graph plots the nonparametric maximum likelihood estimate of $S(t)$ —that is, the probability that a GP will not attempt to raise a new fund by the end of fund quarter t . The number at risk along the x-axis shows the number of funds that have not failed (attempted to fundraise). Sample funds that have not tried to fundraise, or which have not lived long enough for one to observe a fundraising event beyond six years, are excluded from the analysis. The figure indicates that GPs in my sample start fundraising as early as quarter 3, while most of the fundraising occurs between the second and sixth years (which is in line with the results reported by Barber and Yasuda 2017).

—Please see Table 11—

Table 11 reports the estimation results for the hazard rate models of fundraising events as a function of interim performance quartiles for quarter $t-1$. Since I do not want to shut down heterogeneity in performance across fund life by including fixed effects as in prior estimations, I use performance quartiles relative to vintage year cohorts.¹⁵ Columns (1) and (2) report results for funds' performances based on their entire portfolio. The baseline estimates in column (1) show that a fund in the top quartile performance rank has a hazard ratio of 3.525 times that of a fund in the bottom quartile. The hazard ratio of a fund in the second quartile is slightly lower but still statistically significant at the 1% level. Column (2) introduces interaction terms between performance quartiles and the previously used “low reputation dummy” to explore whether interim performance of low reputation funds increases the probability of fundraising. Column (2) shows a significant positive loading on the interaction term, which is to say the probability of raising a fund in quarter t increases more with lagged performance in low reputation funds as compared to high reputation funds. This result also shows that especially low reputation funds

¹⁵In addition, including dummy variables in the models would create the incidental parameters problem typical to non-linear estimators.

cannot systematically repeat their top quartile performance, otherwise being in the top quartile would not have a significantly positive impact on the hazard ratio. The result that low reputation funds have particularly strong incentives to demonstrate either top or second quartile performance to succeed in fundraising is consistent with Barber and Yasuda (2017).

Columns (3-6) break the portfolios into fractions of deals that are unrealized in all quarters prior to failure and into fractions of investments in publicly traded stocks. Based on these fractions, I again calculate performance quartiles. The hazard ratios for being in the top and second quartiles remain statistically significantly positive, and results hold for the interaction with low fund reputation. In line with Barber and Yasuda (2017), I find that top quartile publicly observable performance is very important for fundraising success of low reputation funds (column 6). The hazard ratio increases by 6.1 for these low reputation funds, which is twice as large as the impact of unrealized performance in column (4). This result squares with the findings in Section 5.1 and shows that the verifiability of interim performance is crucial for fundraising success. Because I have been able to use publicly observable performance throughout the holding period, my evidence is consistent with GPs timing fundraising to strong interim performance rather than with reverse causality if performance was only observed from endogenous exits.

6 Conclusion

Private equity reported NAVs have come under increasing scrutiny in recent years. Industry observers and academics have called for regulatory changes to the private equity industry, raising concerns that NAVs are inflated around periods of fundraising, particularly in the case of low reputation funds where abilities have yet to be proven.

Suggestions of regulatory changes are predicated on one possible explanation for the striking evidence of performance peaks at the time a new fund is raised. While it

may be the case that peaks reflect agency problems caused by strategic performance manipulation, this implicit assumption ignores an alternative explanation. That is, general partners can use the heterogeneity in performance across funds and over time as a way of allowing GPs of imperfectly observed quality to showcase their ability. This means instead of strategically manipulating performance estimates, they might time fundraising to true estimates of strong performance. The difficulty of singling out the actual explanation and finding an answer to the question of whether a GP's disclosure can be trusted—in a market of severe information asymmetries—is a data/empirical problem.

Instead of focusing on aggregated valuations on the fund level, this paper makes use of proprietary data on the portfolio company level of U.S. buyout funds from a major institutional investor to show that fund performance peaks are a result of deal composition. That is, current fund peaks stem from strong early deals increasing in value well ahead of fundraising and weak later investments declining in value primarily after fundraising. My overall results show no evidence of inflated valuations on the deal level. In addition, my findings suggest that successful investments reflect some level of skill and poor investments shortly before fundraising are not done because of opportunistic investment behavior.

Consistent with the timing hypothesis as opposed to strategic performance manipulation, GPs appear to time fundraising to true high performance estimates of successful deals, while this performance cannot systematically be repeated in later deals. GPs are more likely to attempt raising a new fund when high NAVs of these successful deals are backed by strong publicly observable performance of stock investments. This result provides an alternative explanation to agency frictions caused by quickening of exits and an incentive to grandstand (Gompers 1996).

The pattern found in my analysis mirrors results on the fund level found in prior studies. At first glance, this finding supports the conjecture that GPs strategically

manipulate performance. That said, my contribution is to provide the first analysis of quarterly reported valuations on the deal level that reveals no evidence of strategic performance manipulation. Regulatory changes with respect to valuation methods in private equity that would seem superficially to be desirable for LPs are not obviously better. My results suggest those investing in a new fund might not get paid what they expected, but can expect what they paid for.

References

- [1] Arcot, S., Fluck, Z., J.-M. Gaspar, and U. Hege, 2015, Fund Managers Under Pressure: Rationale and Determinants of Secondary Buyouts, *Journal of Financial Economics* 115, 102-135.
- [2] Barber, B., and A. Yasuda, 2017, Interim Fund Performance and Fundraising in Private Equity, *Journal of Financial Economics* 124, 172-194.
- [3] Braun, R., T. Jenkinson, and I. Stoff, 2017, How persistent is private equity performance? Evidence from deal-level data, *Journal of Financial Economics* 123, 273-291.
- [4] Brown, G., O. Gredil, and S. Kaplan, 2017, Do Private Equity Funds Game Returns?, Forthcoming in the *Journal of Financial Economics*.
- [5] Chakraborty, I., and M. Ewens, 2017, Managing Performance Signals Through Delay: Evidence From Venture Capital. *Management Science Articles in Advance*, 1-26.
- [6] Gompers, P.A., 1996, Grandstanding in the Venture Capital Industry, *Journal of Financial Economics* 42, 133-56.
- [7] Gompers, P. and J. Lerner, 2000, Money Chasing Deals? The Impact of Fund Inflows on Private Equity Valuations, *Journal of Financial Economics* 55, 281-325.
- [8] Jenkinson, T., M. Sousa, and R. Stucke, 2013, How Fair are the Valuations of Private Equity Funds?, Working Paper, Oxford University.
- [9] Korteweg, A. and M. Sorensen, 2017, Skill and Luck in Private Equity Performance, *Journal of Financial Economics* 124, 535-562.

- [10] Lerner, J., A. Schoar and W. Wongsunwai, 2007, Smart Institutions, Foolish Choices: The Limited Partner Performance Puzzle, *Journal of Finance* 62, 731-764.
- [11] Lopez-de-Silanes, F., L. Phalippou, and O. F. Gottschalg, 2015, Giants at the Gate: Investment Returns and Diseconomies of Scale in Private Equity, *Journal of Financial and Quantitative Analysis* 50, 377-411.
- [12] Metrick, A. and Yasuda, A., 2010, The Economics of Private Equity Funds, *Review of Financial Studies* 23, 2303-41.
- [13] Robinson, D. T. and B. A. Sensoy, 2013. Do Private Equity Fund Managers Earn their Fees? Compensation, Ownership, and Cash Flow Performance, *Review of Financial Studies* 26, 2760-2797.
- [14] Robinson, D. T. and B. A. Sensoy, 2016, Cyclicalities, Performance Measurement, and Cash Flow Liquidity in Private Equity, *Journal of Financial Economics* 122, 521-543.
- [15] Shin, H.-S., 2003, Disclosures and Asset Returns, *Econometrica* 71, 105-133.

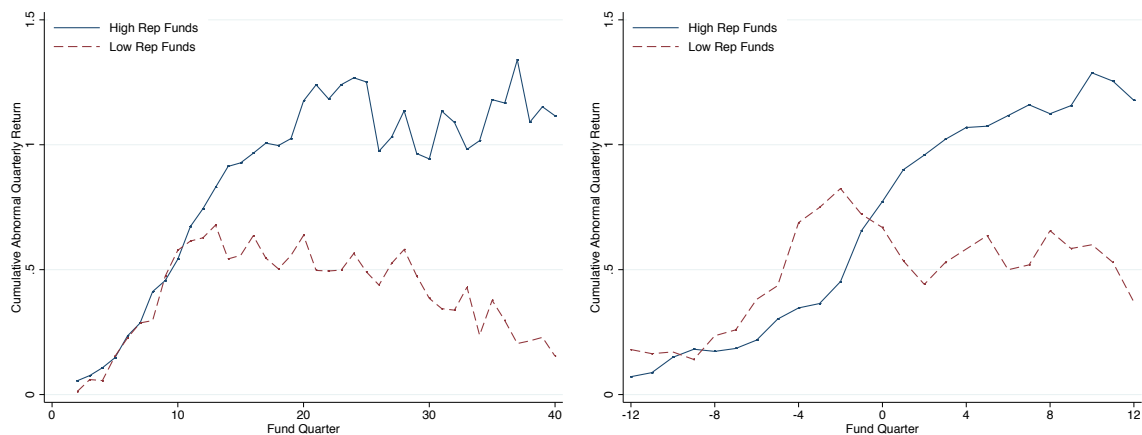


Figure 1: Average cumulative reported fund returns by fund quarter in excess of the cumulative public market returns, $(CAR_t^{NAV} = \prod_{s=1}^t (1 + r_s^{NAV}) - \prod_{s=1}^t (1 + r_s^m))$, with $r_t^{NAV} = (NAV_t + D_t - C_t) / (NAV_{t-1} - 1)$.

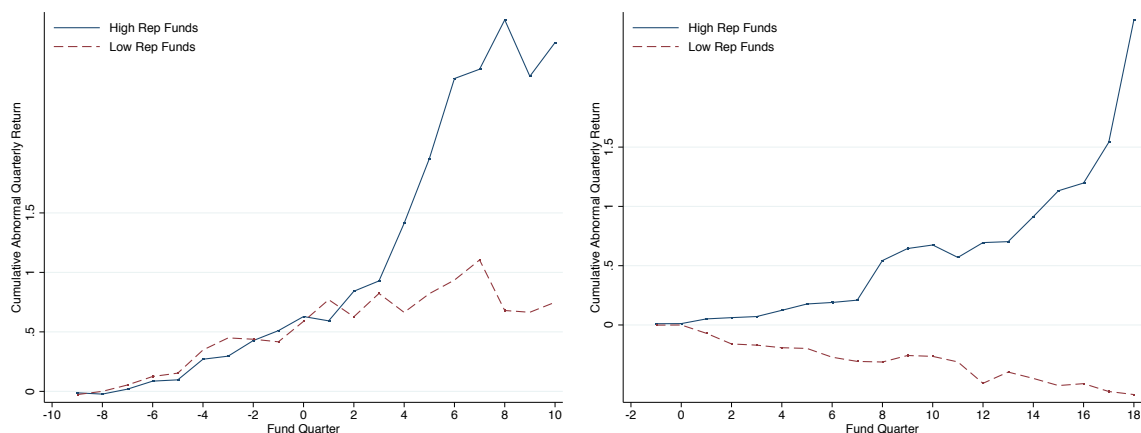


Figure 2: Average cumulative reported fund returns of privately held portfolio companies by fund quarter in excess of the cumulative public market returns, $(CAR_t^{NAV} = \prod_{s=1}^t (1 + r_s^{NAV}) - \prod_{s=1}^t (1 + r_s^m))$, with $r_t^{NAV} = (NAV_t + D_t - C_t) / NAV_{t-1} - 1$. The left (right) hand side graphs display abnormal returns for investments which are made two and a half years (half a year) before the fundraising event quarter and are held in the portfolio for at least one year after the fundraising event.

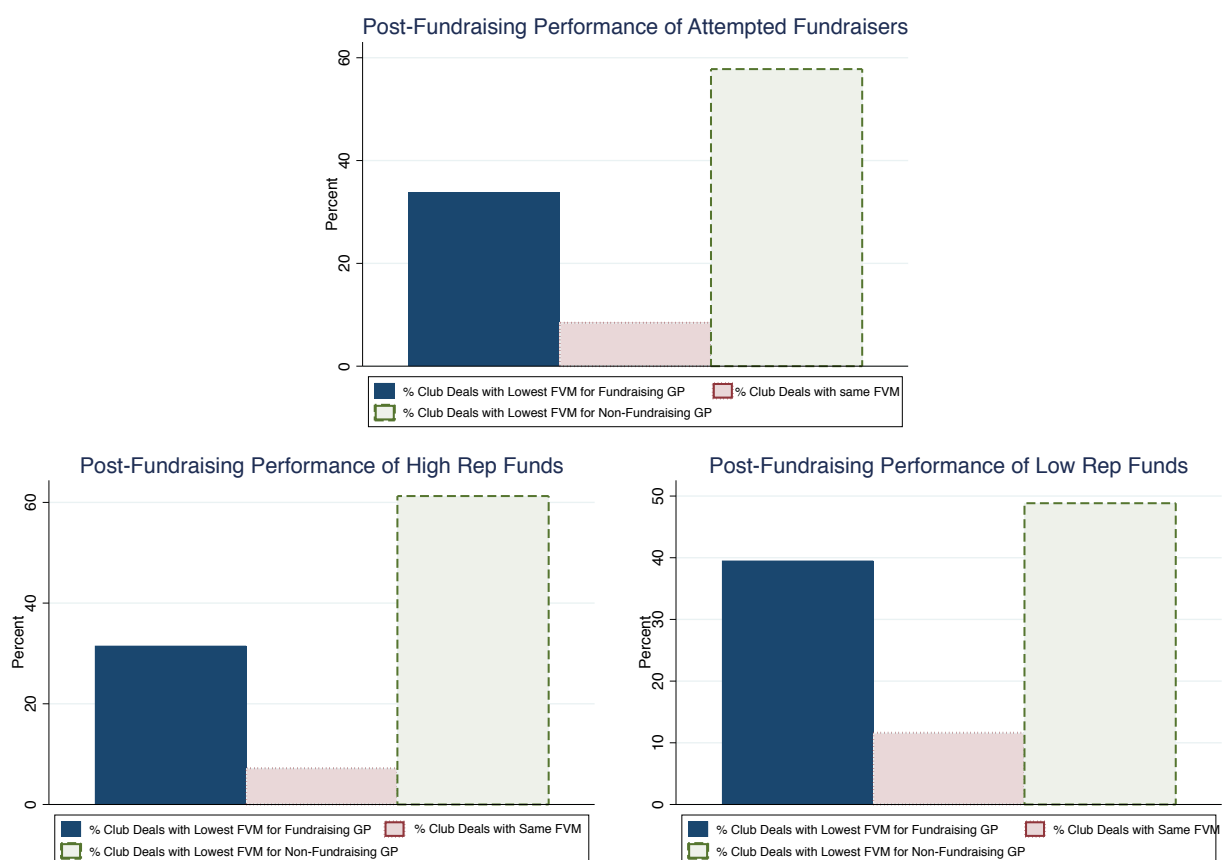


Figure 3: Percent of club deals for which a fundraising GP has the highest/same/lower post-fundraising performance compared to non-fundraising GPs. Post-fundraising performance is based on the value multiple at the time one GP is fundraising (FR), FVM_i ($FVM_i = \frac{\sum_{\tau=FR+1}^T D_{i\tau} + NAV_{iT}}{NAV_{iFR} + \sum_{\tau=FR+1}^T C_{i\tau}}$). Left hand side graph displays club deals of high reputation funds raised by different GPs, while graph on the right hand side shows results for the subsample of low reputation funds raised by different GPs.

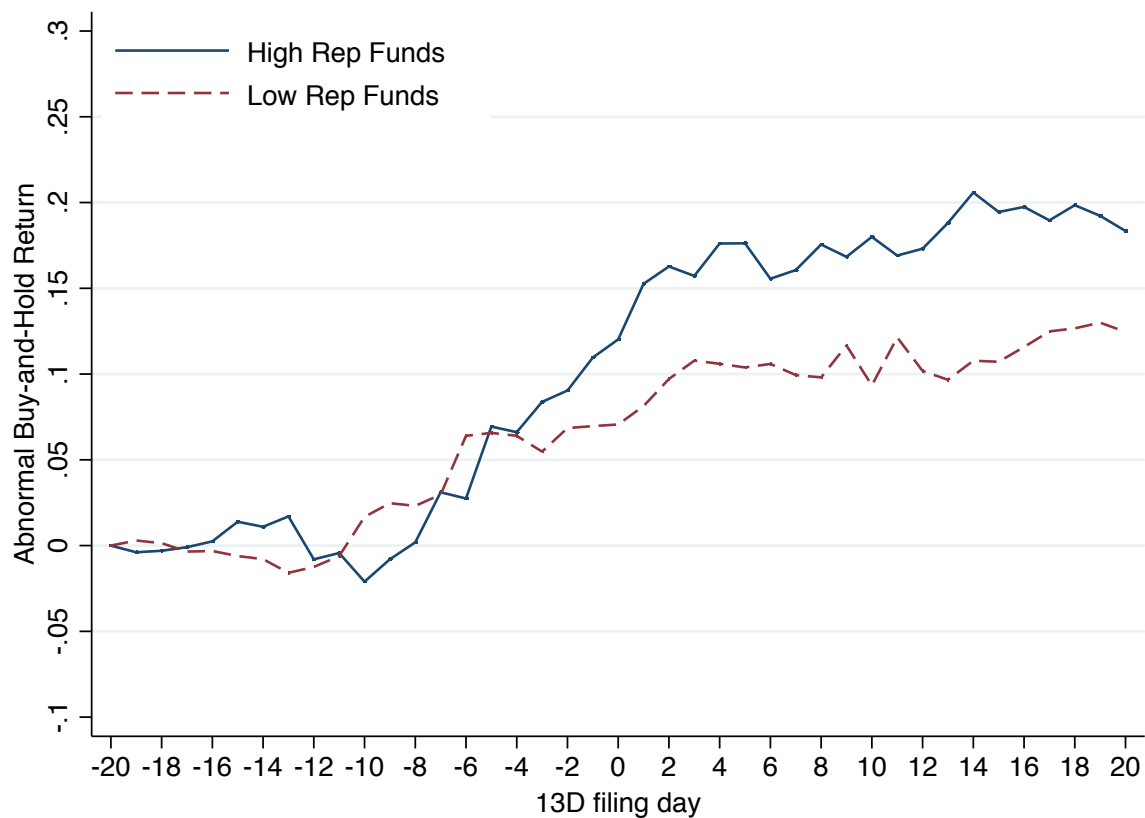


Figure 4: Average buy-and-hold return around the Schedule 13D Filing day, in excess of the buy-and-hold return of the U.S. MSCI industry indices, from 20 days prior the Schedule 13D filing day to 20 days afterwards.

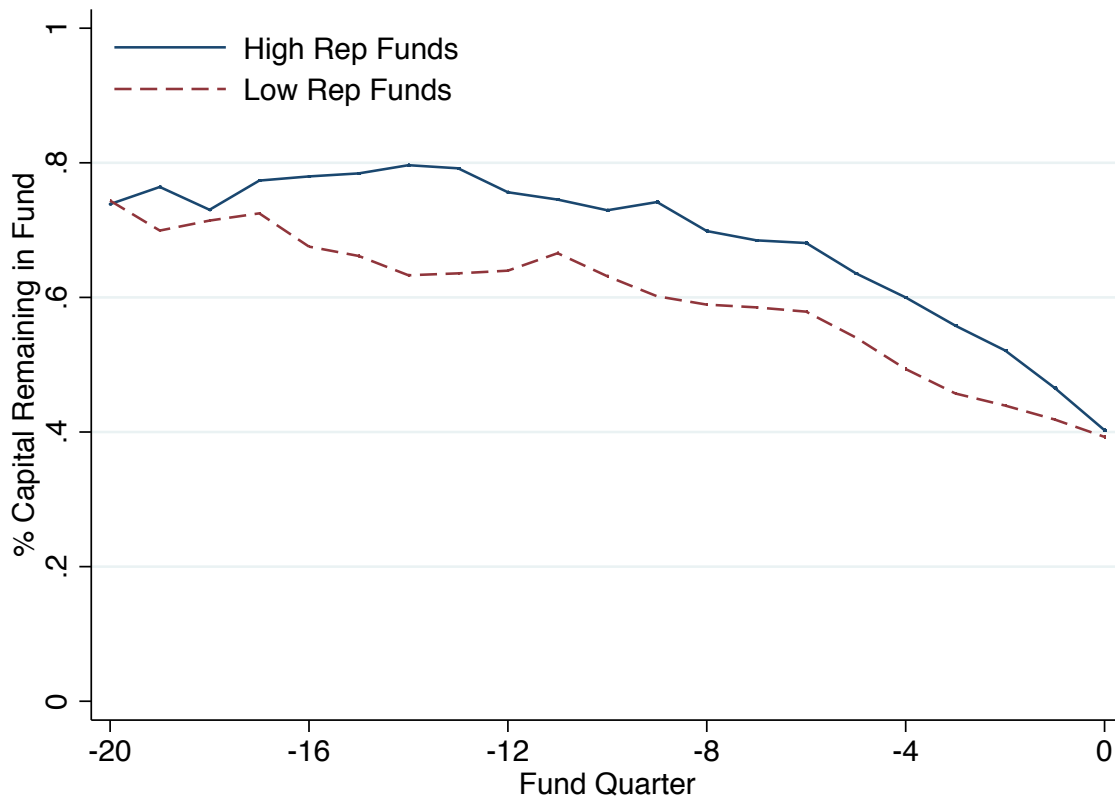


Figure 5: Fraction of capital remaining in the current fund before the closing of the next fund (dry powder). The lines represent the average fractions of remaining capital within either low or high reputation funds, centered at the time of (expected) next fund closing. The two subsamples include all funds that were followed by a(n) (expected) fund closing.

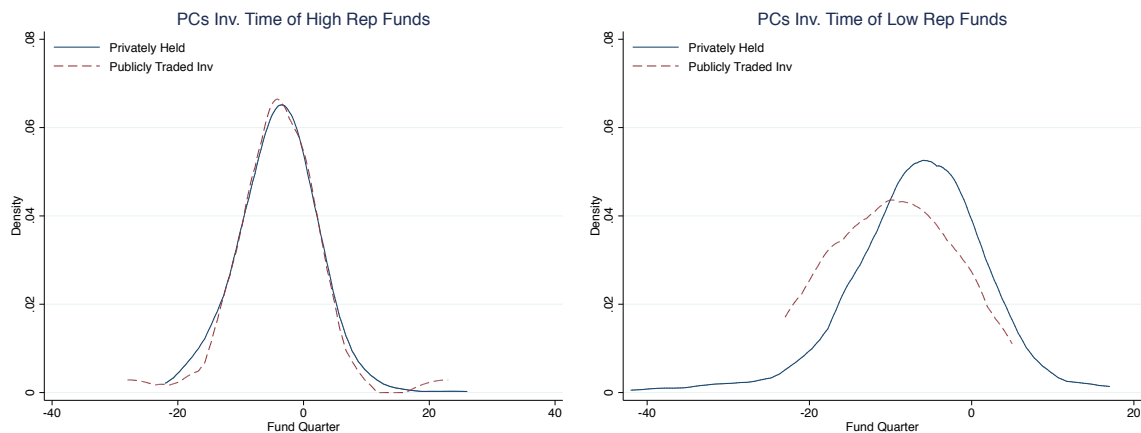


Figure 6: Portfolio companies investment times by fund quarter around the fundraising event (fund quarter 0). The left column of Figure 6 depicts the distribution of investment times for privately held and publicly traded investments of high reputation fundraisers as a function of fund age. This is generated by pooling all initial investments by fund age for each fund type and then plotting the distribution of investments. The right column of Figure 6 depicts the distribution of investment times for funds' privately held and publicly traded investments of low reputation fundraisers.

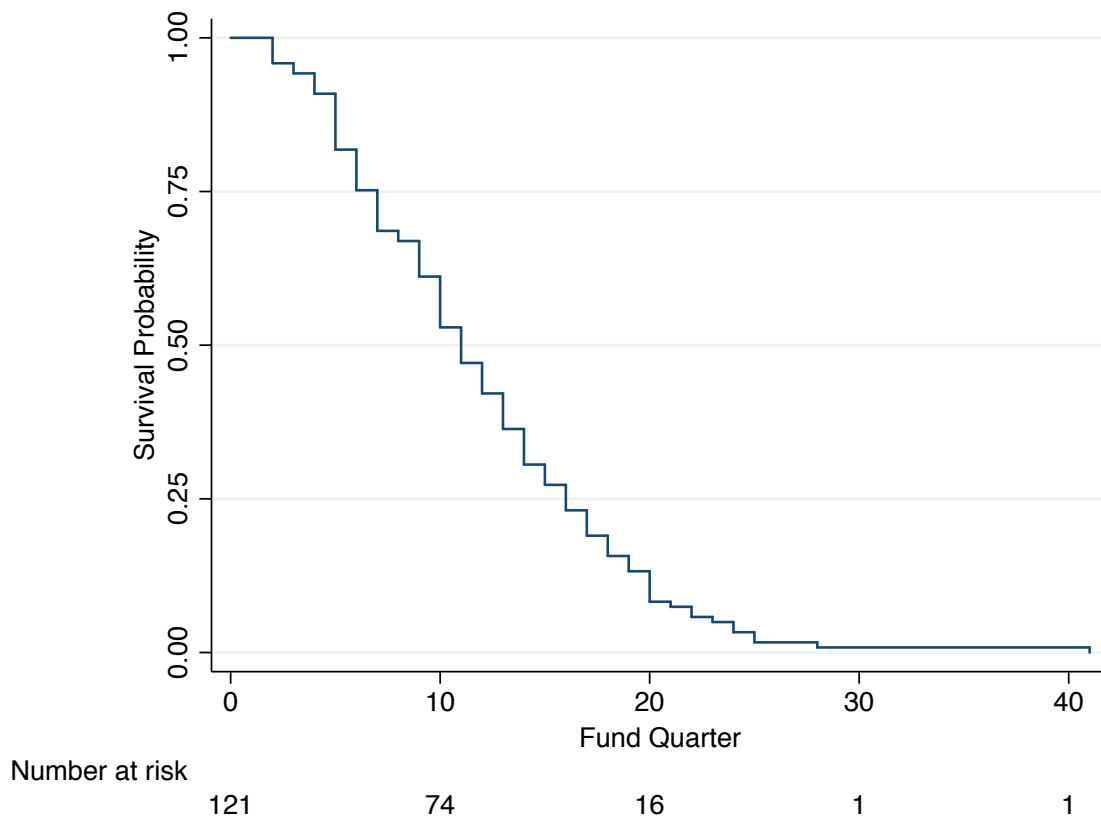


Figure 7: Kaplan-Meier survival graph for fundraising events. This graph shows the probability that a fund does not engage in a fundraising event by fund life quarter. The number at risk displayed at the bottom refers to the number of funds that have yet to raise a follow-on fund.

Table 1: Sample representativeness

This table shows sample representativeness by comparing my dataset to Thomson One, with vintage years between 1996 and 2010. I use Thomson One's Fund and Portfolio Company Statistics with the report date of December 31, 2013. The third column presents the p-values of t-tests and Wilcoxon rank-sum tests (in brackets). "Investments in publicly traded stock" refer to investments that are held publicly traded by the PE fund at least until they are exited. Any investment that is publicly traded at some point of time while held in the portfolio is described as "publicly traded portfolio company". "Yrs. before next fund raised" describes the timing of subsequent buyout funds. "Size of GP" expresses the size of the GPs in questions (across it's previous funds of the last then years) as a fraction of the total capital it raised relative to the total amount raised by all GPs (i.e., investors' commitments) over the ten years preceding each fund; "Age of GP" shows the age of the GP, i.e., the time of the closing of the first partnership that the GP raised to the closing of this fund; "# of past funds" gives the number of past funds of the GP.

	Sample: Fund and portfolio company data	Thompson One: excluding sample data	P-values testing for diff. between sample & Thomson One
<u>Fund Data:</u>			
# of funds	136	1,606	
— if high reputation fund	78		
— if low reputation fund	43		
— if not tried to raise a new fund	15		
# of corresponding GPs	82	780	
<u>Portfolio Company Data:</u>			
# of fund-portfolio-comp pairs	2,779		
— if realized at or above cost	875		
— if realized below cost	584		
— if completely written-off	241		
# of fund-publicly-traded-stock-inv pairs	213		
# of fund-publicly-traded-pc pairs	532		
# of unique portfolio companies	2,400	16,237	
# of unique inv in publicly traded stocks	202	1,310	
# of unique publ traded portfolio comp	387	2,970	
<u>Descriptive Statistics:</u>			
Vintage year	2004.213 (2005)	2003.125 (2004)	0.003 (0.004)
Fund size (in millions)	3106.965 (1500)	579.082 (225)	0.000 (0.000)
Yrs. before next fund raised	3.316 (3)	3.214 (3)	0.600 (0.298)
Size of GP (% of industry \$)	0.023 (0.010)	0.009 (0.001)	0.000 (0.000)
Age of GP (years)	15.485 (12)	10.749 (8)	0.000 (0.000)
# of past funds	4.904 (3)	1.676 (1)	0.000 (0.000)

Table 2: Descriptive statistics on funds holding publicly traded stocks

This table reports the number of funds with investments in publicly traded stocks that are not taken private (first row) and the average number of portfolio companies per fund (second row). The subsequent rows provide descriptive statistics on the fraction of funds' publicly traded stocks relative to the funds' overall portfolio. Means are presented in terms of numbers and invested dollar amount (in brackets). The sample is split up into funds that tried to raise a new fund and those that did not try to raise a new fund (for at least 6 years). For each group means are reported for the entire sample fund life (SFL) and for the time of the fundraising event (FRE).

	$N = 136$		$N = 121$				$N = 15$	
	All sample funds		Attempted Fundraiser				Not tried to raise	
	SFL	FRE	$N = 78$		$N = 43$		SFL	FRE
			High reputation funds	Low reputation funds	SFL	FRE		
# of funds with stock inv.	84	68	52	42	22	17	10	9
# of deals	20.434	14.553	22.564	15.110	16.070	11.154	21.867	18.373
Fund fraction of stocks (% of # [\$])	0.191 [0.263]	0.191 [0.308]	0.201 [0.258]	0.191 [0.294]	0.141 [0.262]	0.155 [0.324]	0.251 [0.287]	0.256 [0.338]
Fund fraction of stock inv. (% of # [\$])	0.077 [0.108]	0.080 [0.122]	0.085 [0.110]	0.086 [0.123]	0.052 [0.109]	0.061 [0.129]	0.094 [0.097]	0.087 [0.103]
Fund fraction of IPOs (% of # [\$])	0.079 [0.105]	0.079 [0.133]	0.075 [0.093]	0.072 [0.125]	0.069 [0.111]	0.070 [0.131]	0.124 [0.145]	0.132 [0.171]
Fund fraction of gone-private transactions (% of # [\$])	0.030 [0.047]	0.026 [0.047]	0.036 [0.051]	0.027 [0.040]	0.017 [0.041]	0.020 [0.060]	0.028 [0.039]	0.031 [0.054]
Fund fraction of gone-private public trans. (% of # [\$])	0.004 [0.004]	0.006 [0.006]	0.005 [0.004]	0.006 [0.006]	0.003 [0.002]	0.004 [0.003]	0.005 [0.007]	0.006 [0.010]

Table 3: Fund-investment pairs across vintage years and industries

This table displays the number of various types of fund-portfolio-company pairs of attempted fundraisers across funds' vintage years and industries.

	# of funds	# of fund deal pairs	# of fund publicly- traded deal pairs	# of fund IPO pairs	# of fund gone- private deal pairs	# of fund gone-private- public deal pairs	# fund private deal pairs
Panel A: Fund-investment pairs across vintage years							
Vintage Year	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1996	1	54	5	5	0	0	44
1997	1	58	5	2	2	0	49
1998	11	220	36	24	11	6	143
1999	8	221	11	22	3	2	183
2000	8	207	16	16	4	3	168
2001	4	139	8	14	2	0	115
2002	3	65	2	1	1	0	61
2003	5	96	6	9	4	0	77
2004	3	38	3	1	0	0	34
2005	18	302	15	33	15	1	238
2006	15	344	19	26	14	0	285
2007	18	361	33	19	6	0	303
2008	15	263	22	7	9	0	225
2009	7	68	3	4	3	0	58
2010	4	15	1	0	0	0	14
Total	121	2,451	185	183	74	12	1,997
Panel B: Fund-investment pairs across industries							
Industry	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Consumer Discretionary	666	48	44	32	3	539	
Consumer Staples	55	8	3	1	1	42	
Energy	60	4	10	0	1	45	
Financials	278	17	23	8	1	229	
Health Care	276	16	21	10	0	229	
Industrials	462	28	30	6	2	396	
Information Technology	481	43	36	17	3	382	
Materials	121	9	13	0	0	99	
Telecommunication Services	39	10	3	0	1	25	
Utilities	13	2	0	0	0	11	
Total	2,451	185	183	74	12	1,997	

Table 4: Performance of Fund-investment pairs across vintage years

This table presents both raw mean and median performance based on the value multiple, VM_i ($VM_i = \sum_{t=0}^T D_{it} + NAV_{iT} / \sum_{t=0}^T C_{it}$) across years of initial investment for each deal. I present investment VMs for the full sample, for realized investments and unrealized investments as of 12/31/2013. The data is winsorized at the 1% extremes.

Investment Year	# of Inv.	Full Sample		Realized		Unrealized	
		(1) Mean	(2) Median	(3) Mean	(4) Median	(5) Mean	(6) Median
1996	14	3.12	1.88	3.12	1.88		
1997	18	2.11	1.89	2.21	1.98	1.28	1.28
1998	30	2.64	1.8	2.74	1.84	1.33	1.33
1999	175	2.12	0.6	2.17	0.55	1.29	0.88
2000	196	1.29	0.29	1.29	0.28	1.26	0.4
2001	85	1.74	0.94	1.83	1.03	0.62	0.52
2002	96	2.45	1.99	2.56	2.19	0.9	0.62
2003	100	2.52	2.1	2.29	2.19	2.25	1.33
2004	157	2.44	2.2	2.6	2.31	1.7	1.73
2005	165	2.44	1.85	2.53	2.16	2.25	1.17
2006	241	1.89	1.47	2.36	1.93	1.49	1.08
2007	351	1.46	1.12	1.79	1.4	1.33	1
2008	246	1.56	1.16	1.58	1.46	1.55	1.05
2009	149	1.97	1.35	1.79	1.48	2.04	1.32
2010	243	1.56	1.19	2	1.93	1.5	1.14
2011	155	1.44	1.16	1.11	0.91	1.45	1.16
2012	14	1.38	1.22			1.38	1.22
2013	16	1.09	1			1.09	1
All deals	2,451	1.83	1.4	2.09	1.61	1.53	1.11

Table 5: Post-fundraising portfolio performance

This table looks at the cross section and presents OLS regression estimates of post-fundraising performance based on the value multiple at the time of a fundraising event (FR), FVM_i ($FVM_i = \sum_{\tau=FR+1}^T D_{i\tau} + NAV_{iT} / NAV_{iFR} + \sum_{\tau=FR+1}^T C_{i\tau}$), for a buyout fund. “Low Rep” is a dummy variable that takes the value of one for a low reputation fund, and zero otherwise. “Unsuccessful public deal pre fundraising” is a dummy variable that takes the value of one for at least a 50% drop in investment return of a publicly traded deal before a new fundraising event, and zero for an investment return larger than -.5 before a fundraising event. “Unsuccessful exit pre fundraising” takes the value of one for at least a 50% drop in value at exit of a private company before a new fundraising event, and zero for a realized return larger than -.5 before a fundraising event. Regression estimates are based on models of calendar year and fund quarter fixed effects (FE). Heteroscedasticity robust standard errors are reported in parentheses below point estimates. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	(1)	(2)	(3)	(4)	(5)
Low Rep	-0.314** (0.150)	-0.648 (0.424)	-0.917* (0.468)	-0.172 (0.316)	-0.131 (0.371)
Unsuccesful public deal pre fundraising		-0.737* (0.376)	-0.894** (0.430)		
Low Rep $\times$ Unsucc. Public			0.97 (1.125)		
Unsuccesful private deal pre fundraising				-0.045 (0.305)	-0.014 (0.397)
Low Rep $\times$ Unsucc. Public					-0.123 (0.604)
Fund Qtr FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.479	0.530	0.536	0.414	0.450
Observations	121	63	63	57	57

Table 6: Post-Fundraising Investment Performance

This table presents OLS regression estimates of post-fundraising performance based on the value multiple at the time of fundraising (FR), $FVM_i = \sum_{\tau=FR+1}^T D_{i\tau} + N \Delta V_{i,T} / N \Delta V_{i,FR} + \sum_{\tau=FR+1}^T C_{i\tau}$, for a buyout fund investment. “Low Rep” is a dummy variable that takes the value of one for a low reputation fund, and zero otherwise. “Deal size” denotes the size of the investment at the time of fundraising. “Unsuccessful public deal pre fundraising” is a dummy variable that takes the value of one for at least a 50% drop in investment return of a publicly traded deal before a new fundraising event, and zero for an investment return larger than -5 before a fundraising event. “Unsuccessful exit pre fundraising” takes the value of one for at least a 50% drop in value at exit of a private company before a new fundraising event, and zero for a realized return larger than -5 before a fundraising event. Regression estimates are based on models of deal quarter, calendar year and industry fixed effects (FE). Standard errors are in parentheses, clustered at the fund level. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Low Rep	-0.085 (0.156)	-0.088 (0.157)	0.278 (0.242)	-0.087 (0.216)	-0.103 (0.230)	-0.419 (0.253)	-0.211 (0.187)	0.176 (0.513)
Deal Size		0.691 (1.053)	-0.571 (0.852)	1.237 (1.964)	1.163 (1.942)	0.937 (1.497)	1.118 (1.542)	-1.053 (2.352)
Deal Size $\times$ Low Rep			4.230 (3.006)					5.332 (5.291)
Unsucc. public deal pre fundraising				-0.253 (0.317)	-0.359 (0.472)			-0.312 (0.486)
Low Rep $\times$ Unsucc. Public					0.237 (0.476)			0.900 (0.666)
Unsucc. exit pre fundraising						0.070 (0.141)	0.176 (0.130)	0.057 (0.207)
Low Rep $\times$ Unsucc. Exit							-0.379 (0.379)	-0.794 (0.490)
Deal Qtr FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.057	0.057	0.06	0.077	0.076	0.056	0.058	0.081
Observations	1,160	1,160	1,160	728	728	663	663	441

Table 7: Performance of Pre-Fundraising Investments

This table presents OLS regression performance estimates based on the value multiple, VM_i ($VM_i = \sum_{t=0}^T D_{it} + NAV_{iT} / \sum_{t=0}^T C_{it}$), for a buyout fund investment. “Shortly before FR” is a dummy variable that takes the value of one for the last two quarters before the first quarter after closing (expected closing) of the next fund (fundraising event), and zero for at least 2 years before the fundraising event. Only investments that remain unrealized for at least one year after the fundraising event are considered. “Deal size” denotes the size of the investment at the time of fundraising. “Unsuccessful public deal pre fundraising” is a dummy variable that takes the value of one for at least a 50% drop in investment return of a publicly traded deal before a new fundraising event, and zero for an investment return larger than -.5 before a fundraising event. Columns 1 and 2 present funds in the sample that went into fundraising, while columns 3 and 4 comprise the subsample of low reputation funds and columns 5 and 6 show results for high reputation funds. “Deal Life FE”, “FR Year FE” and “Industry FE” are fixed effects for the deal life (in quarters), the calendar year at fundraising and the industry. Standard errors are in parentheses, clustered at the fund level. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	Attempted Fundraisers		High Reputation		Low Reputation	
	(1)	(2)	(3)	(4)	(5)	(6)
Inv. shortly before FR	-0.472** (0.210)	-0.701** (0.337)	0.065 (0.193)	-0.191 (0.318)	-1.363*** (0.447)	-1.383** (0.579)
Deal Size	0.826 (0.838)	1.233 (0.921)	0.558 (0.773)	1.204 (1.203)	0.580 (1.874)	0.824 (3.825)
Unsucc. public deal pre fundraising		-0.829 (0.602)		-1.301*** (0.465)		-0.508 (1.409)
Inv. After Fundrais × Unsucc. Public		0.174 (0.778)		1.628** (0.657)		-0.258 (1.247)
Deal Life FE	Yes	Yes	Yes	Yes	Yes	Yes
FR Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.075	0.097	0.068	0.081	0.211	0.349
Observations	708	464	472	326	236	138

Table 8: Returns of Unrealized Investments with Multiple GPs at Fundraising

This table presents OLS regression estimates of post-fundraising performance based on the value multiple at the time of fundraising (FR), FVM_i ($FVM_i = \sum_{\tau=FR+1}^T D_{i\tau} + NAV_{iT} / NAV_{iFR} + \sum_{\tau=FR+1}^T C_{i\tau}$), for a buyout fund investment. Only investments pairs are considered that are held by funds of two or more different GPs. “Event” is a dummy variable that takes the value of one for the first quarter after a GP closed the fundraising period and zero for the GPs without a fundraising event that quarter. Only GP deal pairs are considered if GPs fundraised in different quarters. Otherwise, by construction it cannot be tested if all GPs manipulated performance estimates due to the same fundraising event quarter. “Deal size” denotes the size of the investment at the time of fundraising. “Unsuccessful public deal pre fundraising” is a dummy variable that takes the value of one for at least a 50% drop in investment return of a publicly traded deal before a new fundraising event, and zero for an investment return larger than -.5 before a fundraising event. Columns 1 and 2 present funds in the sample that went into fundraising, while columns 3 and 4 comprise the subsample of low reputation funds and columns 5 and 6 show results for high reputation funds. Regression estimates are based on models of deal life, calendar year and industry fixed effects (FE). Standard errors are in parentheses, clustered at the fund level. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	Attempted Fundraisers		High Reputation		Low Reputation	
	(1)	(2)	(3)	(4)	(5)	(6)
Unrealized Privately Held Investments That Are Made Before Fund Raising Event						
Event	0.031 (0.148)	0.152 (0.182)	0.046 (0.174)	0.096 (0.222)	0.031 (0.273)	0.248 (0.358)
Deal Size		-5.035 (3.769)		-1.291 (2.165)		-5.424 (4.308)
Unsucc. public deal pre fundraising		-0.405 (0.264)		-0.499* (0.286)		0.594 (0.749)
Deal Life FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.117	0.155	0.186	0.251	0.033	0.124
Observations	443	319	335	230	108	89

Table 9: Dry powder pre-fundraising

This table presents OLS regression estimates of capital available for new investments (dry powder). The dependent variable, dry powder, is measured as the difference between the sum of capital invested by a buyout fund over time and the total committed capital, divided by total committed capital. “Shortly before FR” is a dummy variable that takes the value of one for the last two quarters before the first quarter after closing (expected closing) of the next fund (fundraising event), and zero otherwise. “Low Rep” is a dummy variable that takes the value of one for a low reputation fund, and zero otherwise. “Unsuccessful public deal pre fundraising” is a dummy variable that takes the value of one for at least a 50% drop in investment return of a publicly traded deal before a new fundraising event, and zero for an investment return larger than -.5 before a fundraising event. “Unsuccessful exit pre fundraising” takes the value of one for at least a 50% drop in value at exit of a private company before a new fundraising event, and zero for a realized return larger than -.5 before a fundraising event. “FR Qtr FE” and “FR Year FE” are fixed effects for the calendar year and the fund quarter at fundraising. Standard errors are in parentheses, clustered at the fund level. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	(1)	(2)	(3)	(4)
Shortly before FR	-0.240*** (0.012)	-0.256*** (0.015)	-0.259*** (0.016)	-0.255*** (0.020)
Ln(fund size)	0.030** (0.012)	0.029** (0.012)	0.029** (0.012)	0.027** (0.012)
Low Rep		-0.041 (0.033)	-0.043 (0.032)	-0.028 (0.033)
Low Rep $\times$ Shortly before FR		0.043 (0.030)	0.043 (0.032)	0.038 (0.035)
Unsucc. public deal pre fundraising			-0.288*** (0.037)	
Low Rep $\times$ Unsucc. Public			0.018 (0.078)	
Shortly before FR $\times$ Unsucc. Public			0.262*** (0.029)	
Low Rep $\times$ Shortly before FR $\times$ Unsucc. Public			-0.091 (0.059)	
Unsucc. exit pre fundraising				-0.095*** (0.031)
Low Rep $\times$ Unsucc. Exit				-0.181** (0.082)
Shortly before FR $\times$ Unsucc. Exit				0.060 (0.037)
Low Rep $\times$ Shortly before FR $\times$ Unsucc. Exit				0.071 (0.067)
FR Qtr FE	Yes	Yes	Yes	Yes
FR Year FE	Yes	Yes	Yes	Yes
Adj. R ²	0.472	0.474	0.487	0.517
Observations	1,404	1,404	1,404	1,404

Table 10: Investment Performance of (Un)attempted Fundraisers

This table presents OLS regression of performance estimates based on the value multiple, VM_i ($VM_i = \sum_{t=0}^T D_{it} + NAV_{iT} / \sum_{t=0}^T C_{it}$), for a buyout fund investment. Columns 1 and 2 present funds in the sample that went into fundraising, while columns 3 and 4 comprise funds that did not try to raise a new fund. The last columns (5 and 6) subsume both samples of attempted and unattempted fundraisers. Only low reputation funds are considered. “Early Inv.” is a dummy variable that takes the value of one for investments that have been initiated in the first two fund life years, and zero otherwise. “Deal size” denotes the size of the investment at the time of fundraising. Regression estimates are based on models of deal life, industry and fund vintage year fixed effects (FE). Standard errors are in parentheses, clustered at the fund level. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	Low Reputation Funds					
	Attempted Fundraisers (AF)		Unattempted Fundraisers (UF)		AF + UF	
	(1)	(2)	(4)	(5)	(7)	(8)
Early Inv.	0.900** (0.391)	0.908** (0.393)	-0.228 (0.622)	-0.219 (0.643)	0.601 (0.370)	0.613 (0.371)
Deal Size		0.927 (1.041)		1.819 (4.189)		1.242 (1.016)
Deal Life FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Vintage Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.072	0.071	0.003	0.004	0.054	0.054
Observations	691	691	159	159	850	850

Table 11: The Impact of Interim Performance Ranking on Fundraising

This table presents hazard rate models of fundraising events as a function of interim performance quartiles. I estimate Cox proportional hazards models using quarterly observations of quartile performance ranks based on the sample's value multiples. Hazard ratios are presented, which can be easily converted into coefficients. The failure event is the first quarter after closing (expected closing) of the next fund (fundraising event) so that each fund is at risk during the pre-event period. I allow the failure to occur anytime in the fund's life. I only include funds that actually attempted to raise a follow-on fund. Columns 1 and 2 consider all investments by funds, whereas columns 3 and 4 report results for portfolios of unrealized investments as of period t and columns 5 and 6 display portfolios in publicly traded stock investments (this fraction is benchmarked to the funds' interim performance based on all their investments up to that date). Top (second/third) quartile dummy variable takes a value of one if the quarter $t-1$ performance rank is in the top (second/third) quartile among its vintage-year cohort (either based on all investments or only on unrealized investments). "Low Rep" is a dummy variable that takes the value of one for a low reputation fund, and zero otherwise. The last three explanatory variables in models 2, 4 and 6 are interaction terms between the "Low Rep" dummy and the quartile performance rank dummies. Standard errors are in parentheses, clustered at the fund level. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	time-varying?	Funds consisting of		Funds Fraction in			
		All Investments		Unrealized Investments		Inv in Publ Traded Stocks	
		(1)	(2)	(3)	(4)	(5)	(6)
Top Quartile	yes	3.525*** (1.099)	2.125* (0.860)	3.360*** (1.087)	2.128* (0.872)	3.764*** (1.171)	2.466** (0.920)
2 nd Quartile	yes	3.044*** (0.940)	2.193** (0.868)	2.393*** (0.752)	2.394** (0.954)	2.778*** (1.099)	2.585** (1.060)
3 rd Quartile	yes	2.018** (0.635)	1.801 (0.743)	1.590 (0.536)	1.708 (0.747)	1.598 (0.808)	1.448 (0.894)
Ln(fund size)	no	1.011 (0.046)	0.993 (0.047)	1.065 (0.049)	1.065 (0.057)	0.933 (0.076)	0.912 (0.082)
Low Rep	no		0.414* (0.204)		0.544 (0.302)		0.518 (0.265)
Top $\times$ Low Rep	yes		3.248** (1.925)		3.141* (2.048)		6.103** (4.491)
2 nd $\times$ Low Rep	yes		1.960 (1.169)		0.857 (0.554)		1.374 (1.322)
3 rd $\times$ Low Rep	yes		1.265 (0.775)		0.802 (0.535)		1.550 (1.546)
LR test: all coeff. = 0 (χ^2)		20.48***	26.95***	20.67***	26.72***	21.00***	33.39***
# fund quarters		1,404	1,404	1,404	1,404	559	559
# funds		121	121	121	121	68	68

Appendix: Results Based On IRR

Table A1: Performance of Fund-investment pairs across vintage years

This table presents both raw mean and median performance based on internal rates of returns (IRRs) of individual portfolio investments across years of initial investment. I present investment IRRs for the full sample, for realized investments and unrealized investments as of 12/31/2013. The data is winsorized at the 1% extremes.

Investment Year	Full Sample		Realized		Unrealized	
	Panel A: Raw performance based on IRRs					
	(1) Mean	(2) Median	(3) Mean	(4) Median	(5) Mean	(6) Median
1996	0.44	0.14	0.44	0.14		
1997	0.38	0.19	0.46	0.43	0.03	0.03
1998	0.27	0.07	0.28	0.06	0.07	0.07
1999	0.15	0.06	0.17	0.07	-0.06	-0.07
2000	0.12	0.07	0.12	0.06	0.13	0.11
2001	0.30	0.30	0.32	0.32	-0.17	-0.17
2002	0.39	0.29	0.4	0.3	0.08	0.08
2003	0.38	0.27	0.37	0.27	0.5	0.2
2004	0.42	0.31	0.49	0.35	0.08	0.03
2005	0.30	0.21	0.38	0.25	0.02	-0.02
2006	0.21	0.14	0.33	0.22	-0.02	-0.01
2007	0.10	0.10	0.22	0.18	-0.07	-0.09
2008	0.19	0.14	0.35	0.31	0.01	-0.04
2009	0.37	0.33	0.66	0.61	0.1	0.05
2010	0.20	0.22	0.33	0.34	0.08	0
2011	-0.03	-0.08	0.21	0.17	-0.1	-0.09
2012	0.04	0.02			0.04	0.02
2013	0.01	0			0.01	0
All deals	0.21	0.16	0.32	0.22	0.01	0

Table A2: Performance of Pre-Fundraising Investments

This table presents OLS regression performance estimates based on the internal return (IRR), for a buyout fund investment. “Shortly before FR” is a dummy variable that takes the value of one for the last two quarters before the first quarter after closing (expected closing) of the next fund (fundraising event), and zero for at least 2 years before the fundraising event. All considered investments remained unrealized at least one year after the event. “Deal size” denotes the size of the investment at the time of fundraising. “Unsuccessful public deal pre fundraising” is a dummy variable that takes the value of one for at least a 50% drop in investment return of a publicly traded deal before a new fundraising event, and zero with less than a 50% drop before a new fundraising event. Columns 1 and 2 present funds in the sample that went into fundraising, while columns 3 and 4 comprise the subsample of low reputation funds and columns 5 and 6 show results for high reputation funds. “Deal Life FE”, “FR Year FE” and “Industry FE” are fixed effects for the deal life (in quarters), the calendar year at fundraising and the industry. Standard errors are in parentheses, clustered at the fund level. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	Attempted Fundraisers		High Reputation		Low Reputation	
	(1)	(2)	(3)	(4)	(5)	(6)
Inv. shortly before FR	-0.059 (0.041)	-0.063 (0.054)	-0.026 (0.047)	-0.007 (0.069)	-0.261** (0.128)	-0.196** (0.095)
Deal Size	0.170 (0.165)	0.055 (0.225)	0.171 (0.220)	-0.240 (0.297)	0.279 (0.344)	-0.927 (1.793)
Unsucc. public deal pre fundraising		0.035 (0.070)		0.031 (0.127)		0.495 (0.506)
Inv. shortly before FR × Unsucc. public		-0.118 (0.099)		-0.188 (0.181)		-0.182 (0.997)
Deal Life FE	Yes	Yes	Yes	Yes	Yes	Yes
FR Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.068	0.075	0.057	0.060	0.118	0.152
Observations	708	464	472	326	236	138

Table A3: Investment Performance of (Un)attempted Fundraisers

This table presents OLS regression of performance estimates based on the internal return (IRR), for a buyout fund investment. Columns 1 and 2 present funds in the sample that went into fundraising, while columns 3 and 4 comprise funds that did not try to raise a new fund. The last columns (5 and 6) subsume both samples of attempted and unattempted fundraisers. Only low reputation funds are considered. “Early Inv.” is a dummy variable that takes the value of one for investments that have been initiated in the first two fund life years, and zero otherwise. “Deal size” denotes the size of the investment at the time of fundraising. Regression estimates are based on models of deal life, industry and fund vintage year fixed effects (FE). Standard errors are in parentheses, clustered at the fund level. One/ two/ three asterisks represent two-tailed significance at a 10%/ 5%/1% level, respectively.

	Low Reputation Funds					
	Attempted Fundraisers (AF)		Unattempted Fundraisers (UF)		AF + UF	
	(1)	(2)	(4)	(5)	(7)	(8)
Early Inv.	0.074** (0.035)	0.072* (0.038)	-0.137 (0.132)	-0.139 (0.134)	0.05 (0.044)	0.048 (0.046)
Deal Size		-0.118 (0.288)		-0.134 (0.217)		-0.092 (0.165)
Deal Life FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Vintage Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.062	0.06	0.028	0.027	0.063	0.061
Observations	691	691	159	159	850	850